

THE HONG KONG UNIVERSITY OF SCIENCE & TECHNOLOGY (GUANGZHOU)

Deep Learning | HA3: Diffusion Model Experiments on MNIST

Due Date: [See web](#)

Instructions:

- Your answers should be typed, not handwritten.
- You may submit PDF or DOCX format.
- Submissions must be made via Canvas.

Copyright Statement: The materials provided by the instructor in this course are for the use of the students enrolled in the course. Copyrighted course materials may not be further disseminated.

Objective

In this assignment, you will extend the diffusion model implementation from the tutorial and conduct a series of experiments on the **MNIST dataset**. The goal is to analyze how different design choices affect the quality of generated images.

Assignment Details

1. Baseline Setup

- Start from the diffusion model implementation provided in the tutorial.
- Use MNIST as the dataset.

2. Experimental Variations

You are required to perform experiments by varying the following components:

- **Noise Schedule:** Try at least two different noise schedules.
- **Diffusion Time Steps:** Use at least two different numbers of time steps.
- **Sampling Method:** Compare **DDPM** and **DDIM**.

Requirement: You must conduct at least **8 experiments** in total.

3. Evaluation Metrics

To evaluate the quality of generated images, consider using:

- **Inception Score (IS)**
- **Fréchet Inception Distance (FID)**

4. Additional Analysis

Diffusion models map real data to a unit Gaussian distribution in latent space.

- Demonstrate that starting from latent vectors far from the origin leads to poor image quality.
- Use your **best trained model** to illustrate this phenomenon.

Reporting

Your report should be well-organized and include:

1. **Summary:** Briefly describe your setup and experiment configurations.
2. **Methodology:** Explain how you varied the noise schedule, time steps, and sampling methods.
3. **Results:** Present the results of all experiments and compare them.
4. **Analysis:** Provide insights into how different factors affect performance.
5. **Visualization:** Include generated image samples and comparisons across different settings.

Submission

- Submit your report via Canvas.
- No code submission is required.
- Similarity scores will be computed.

Academic Integrity

All submissions must follow the course policy on collaboration and citation. Properly acknowledge any external resources used.